$$\begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 1 & 3 & 0 & 2 \end{bmatrix} \xrightarrow[\ -\frac{1}{5}\times \text{Row}1]{\text{Row}3=\text{Row}3} \begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 7/5 & -7/5 & 8/5 \end{bmatrix} \xrightarrow[\times 5]{\text{Row}3=\text{Row}3}$$

$$\begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 7 & -7 & 8 \end{bmatrix} \xrightarrow[\times 1/7]{\text{Row}3=\text{Row}3} \begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 1 & -1 & 8/7 \end{bmatrix} \xrightarrow[-\text{Row}2]{\text{Row}3=\text{Row}3} \begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 29/7 \end{bmatrix}$$

Verification of whether the final matrix is in echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that all non-zero rows must be above all zero rows is vacuously True.

2. Row 1 Leading element $= 5$ in Column 1
   Row 2 Leading element $= 1$ in Column 2
   Row 3 Leading element $= 29/7$ in Column 4
   $\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True.

3. Row 2 Column 1 $=$ Row 3 Column 1 $= 0$
   Row 3 Column 2 $= 0$
   $\therefore$ The condition that for each pivot column, all the elements below the pivot position must be 0 is True.

So, the matrix $\begin{bmatrix} 5 & 8 & 7 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 29/7 \end{bmatrix}$ is in echelon form.

There is a row in the matrix $(0\ 0\ 0\ 29/7)$ of the form $[0\ 0\ 0 \ldots\ 0\ b]$, where $b \neq 0$. By Theorem 2, the corresponding system of equations is inconsistent, i.e. there exists no solution set.

$\boxed{\text{So, } u \notin \text{Span}\{a_1, a_2, a_3\}}$ □